

When Clinicians Reshape the Model: A Protocol for a Retrospective Design Case Study of Expert Feedback in a Data Science Capstone

Yeoeun Choi

Independent Researcher, Boston, MA, USA

ychoi0908@gmail.com

Study protocol. No results yet. Not peer reviewed.

Abstract

Human-computer interaction research argues that the people closest to a problem should shape a model's design early, but first-hand accounts of a data science team changing course because of practitioners are rare. In a graduate capstone, my four-person team built decision-support tools for an early intervention provider serving young children, consolidating 14 years of scattered operational records. A pilot that used a large language model to parse therapists' free-text availability failed in documented ways, and after staff feedback the team shifted from backend automation to clearer dashboards with the comparison and filtering views clinicians asked for. This protocol describes a retrospective design case study of that turn. I will reconstruct a dated history of design decisions from project artifacts, mark each as team-initiated or practitioner-initiated, and ask which practitioner concepts were missing from the original plan and what gathering them earlier would have changed. The study uses existing artifacts only, involves no new participants, and will not begin until the client and my capstone advisor approve publication.

Keywords: human-AI interaction; design case study; domain experts; decision support; dashboards; early intervention

1. Introduction

Data science teams are often trained to treat a project as a modeling problem: find the data, build the pipeline, report the metrics. In my capstone, that framing broke down. The tool mattered only if the staff using it could understand and trust it, and they told us so after we had already built much of the backend.

This protocol plans a short case study of that experience. Its aim is not to evaluate the capstone but to use it as a concrete, documented instance of a question that HCI research has raised mostly in general terms: when should domain experts shape a data system's design, and what is lost when their input arrives late?

What changes when the people who understand the problem best shape a data system only after it has been built?

2. Background

Yang et al. (2020) identify two sources of the distinctive difficulty of designing human-AI interaction: uncertainty about what an AI system can do, and the complexity of its outputs. Amershi et al. (2019) propose 18 guidelines for human-AI interaction, validated with 49 design practitioners against 20 AI-infused products. Lam et al. (2023) argue that early model design should center on concepts that matter to the problem rather than on implementation details. Table 1 summarizes what this case can add to each.

Table 1. Prior work and what this case adds.

Work	Claim	How this case adds to it
Yang et al. (2020)	AI is hard to design around because	A concrete instance: the parsing

	its capabilities are uncertain	pilot's failures changed the design target
Amershi et al. (2019)	18 guidelines for human-AI interaction	Shows which guidelines a non-HCI team reached on its own, and how late
Lam et al. (2023)	Center early model design on concepts that matter	Shows the cost when those concepts arrive as feedback after the build

3. Research questions

1. **RQ1.** At which points in the project did staff and clinician input change the design, and what did each change?
2. **RQ2.** Which concepts that clinicians cared about, such as the comparisons and filters they requested, were absent from the team's original plan?
3. **RQ3.** If those concepts had been gathered before the build, which work would have been avoided or done differently?

4. Method

4.1 The case

The case is my Carnegie Mellon MSBA capstone (2025). A four-person team worked with an early intervention provider to consolidate 14 years of operational records into decision-support dashboards under HIPAA constraints. I co-led data analysis and dashboard development. The team piloted a large language model to parse therapists' free-text availability, documented where it failed, and designed a manual verification fallback. After staff feedback, the deliverable shifted toward dashboard clarity and the comparison and filtering views clinicians requested.

4.2 Permission and privacy

No case-study writing begins until the following are complete. If permission is refused, the study will be reduced to a methods reflection with no client details, or dropped.

- Reread the capstone agreement for publication and confidentiality terms.
- Ask my capstone advisor whether a de-identified case study is permitted.
- Ask the client for written permission and offer to share the draft before posting.
- Ask each teammate whether they want to be named or acknowledged.
- Refer to the client only by a generic description, such as "a regional early intervention provider."

4.3 Sources

The design history will be reconstructed from dated artifacts rather than from memory (Table 2).

Table 2. Sources and how each will be handled.

Source	What it shows	Handling
Proposal and scope documents	The original plan and success criteria	Quote goals only, no client data
Meeting notes and emails with staff	When feedback arrived and what it asked for	Paraphrase; no staff names
Parsing pilot logs and failure notes	Where automation broke down	Report failure types, never record content
Dashboard versions and screenshots	How the design changed	Redraw as wireframes with synthetic data
Final report and contributions summary	What was delivered and who did what	Credit teammates accurately

4.4 Procedure

1. Build a dated timeline of every design decision from the artifacts.
2. Mark each decision as team-initiated or practitioner-initiated.
3. For each practitioner-initiated change, record the concept it introduced, such as a comparison, a filter, or a trust concern.
4. For each such change, write a short counterfactual: what an early concept-gathering step would have changed.

5. Analysis and rigor

The analysis maps each design change onto the guidelines of Amershi et al. (2019) and the argument of Lam et al. (2023), and states plainly where the case does and does not fit them.

- Every claim about timing cites a dated artifact; claims from memory are labeled as recollection.
- At least one teammate and my capstone advisor will review the draft to confirm the account.
- No patient, family, or therapist data will appear; all figures will use synthetic data.

6. Limitations

- One project and one author's view: the case illustrates, it does not generalize.
- The account is retrospective, not observational.
- I co-led data analysis and dashboards but did not lead the team, so some decisions are seen secondhand.

7. Declarations

Origin of this work. The case is my Carnegie Mellon MSBA capstone project (2025), completed with three teammates.

Use of AI tools. I used Claude (Anthropic), an AI assistant, to verify references against Crossref records and to draft text and tables.

Funding. None.

Competing interests. None.

References

- Amershi, S., Weld, D., Vorvoreanu, M., Fourney, A., Nushi, B., Collisson, P., Suh, J., Iqbal, S., Bennett, P. N., Inkpen, K., Teevan, J., Kikin-Gil, R., & Horvitz, E. (2019). Guidelines for human-AI interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (pp. 1–13). ACM. <https://doi.org/10.1145/3290605.3300233>
- Lam, M. S., Ma, Z., Li, A., Freitas, I., Wang, D., Landay, J. A., & Bernstein, M. S. (2023). Model sketching: Centering concepts in early-stage machine learning model design. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (pp. 1–24). ACM. <https://doi.org/10.1145/3544548.3581290>
- Yang, Q., Steinfeld, A., Rosé, C., & Zimmerman, J. (2020). Re-examining whether, why, and how human-AI interaction is uniquely difficult to design. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (pp. 1–13). ACM. <https://doi.org/10.1145/3313831.3376301>